An Intelligent Cloud-Based Disaster Management System with Multimodal Disaster Prediction, Quantum-Inspired Resource Allocation, and Blockchain-Based Tracking

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***Abstract*—Disaster management requires seamless coordination between disaster identification, impact assessment, emergency resource estimation, administrative authorization, multi-warehouse allocation, and movement tracking. In conventional operational workflows, these activities are handled across fragmented, siloed systems, leading to delayed response times and lack of end-to-end traceability. This paper presents an intelligent cloud-based disaster management system that integrates predictive disaster analytics, multimodal impact assessment, standard humanitarian resource estimation, hierarchical state-national approval workflows, quantum-inspired multi-warehouse optimization, and blockchain-based audit logging within a unified architecture. The system ingests historical disaster records, real-time meteorological observations, and Copernicus Sentinel satellite imagery to detect hazard risks. When potential crises are identified, automated requirement rules compute itemized emergency demands, initiating an administrative authorization chain. Upon approval, a Quantum Approximate Optimization Algorithm (QAOA) formulation running on a parameterized variational circuit solves the constrained multi-warehouse allocation problem, balancing geodesic Haversine transit distances, inventory availability, and critical resource priorities. Allocated supplies are tracked across dispatch, transit, arrival, and delivery stages, with cryptographic state transitions anchored to an immutable Ethereum smart contract. The system is implemented using React, Node.js, Express, MySQL, FastAPI, Qiskit, and Solidity. Empirical evaluation demonstrates reliable flood prediction (up to 93% confidence), sub-second quantum optimization convergence, two-phase atomic stock reservation, and tamper-proof audit trails.**

***Index Terms*—Disaster Management, Cloud Computing, Machine Learning, Satellite Remote Sensing, Sentinel-1, Resource Allocation, Quantum-Inspired Optimization, QAOA, Blockchain, Smart Contracts, Humanitarian Logistics.**

I. Introduction

Natural disasters such as floods, cyclones, landslides, earthquakes, and severe meteorological anomalies inflict devastating socio-economic disruptions, displacement of populations, and catastrophic damage to critical infrastructure. Effective emergency response demands rapid, reliable situational intelligence and synchronized operational execution spanning hazard detection, population impact assessment, relief supply calculation, multi-echelon government authorization, warehouse dispatch, and continuous transit monitoring. Recent research has investigated emerging computing paradigms for disaster response, including unmanned aerial vehicle (UAV)-assisted edge computing [1], blockchain-based resilience networks [2], post-disaster communication frameworks [3], quantum-adaptive resource allocation [4], and digital twins for emergency monitoring [5].

A persistent operational vulnerability in disaster logistics is the structural fragmentation of information flows. Disaster intelligence generated by meteorological agencies is rarely

connected directly to supply-chain inventories or distribution systems. Consequently, blockchain and distributed ledger technologies have been explored to enhance multi-agency coordination, transparency, and aid traceability [6]–[8]. Recent bibliometric and architectural surveys underscore the necessity of decentralized verification in emergency management [9], [10], while specialized frameworks have examined distributed ledgers for seismic mitigation, disaster prevention, and relief operations [11], [12].

Concurrently, the increasing availability of open-access Earth Observation (EO) data offers unprecedented spatial situational awareness. Copernicus Sentinel-1 Synthetic Aperture Radar (SAR) observations are especially vital for flood mapping because microwave radar penetrates dense cloud cover and operates independently of daylight illumination [13], [14]. Satellite observations have also been combined with advanced computing paradigms to model complex spatial vulnerabilities [15].

Emergency resource distribution represents a mathematically demanding NP-hard combinatorial problem [16]. Relief kits must be sourced from geographically distributed warehouses subject to supply constraints, travel time, and disaster severity. Quantum and quantum-inspired optimization algorithms, particularly the Quantum Approximate Optimization Algorithm (QAOA) [22], have emerged as promising frameworks for solving constrained combinatorial optimization problems [1], [4]. Furthermore, blockchain-based audit trails ensure that dispatched humanitarian assets are tracked transparently without risk of diversion or tampering [17]–[21].

To address these interconnected challenges, this paper presents a unified, cloud-native national disaster management and relief distribution platform. The proposed platform couples multimodal disaster intelligence with an end-to-end operational pipeline encompassing:

1) Spatiotemporal disaster prediction using an ensemble of 100 decision trees trained on multi-decade climate and disaster event datasets;
2) Remote sensing inundation analysis utilizing Sentinel-2 optical water indices and Sentinel-1 SAR backscatter calibration;
3) Calibrated severity scoring and automated per-capita humanitarian relief demand estimation;
4) Geodesic Haversine distance-weighted multi-warehouse allocation solved via QAOA variational quantum circuits;
5) A two-phase atomic inventory reservation mechanism preventing concurrent double-allocation;
6) Multi-echelon administrative approval workflows and cryptographically sealed Ethereum smart contract audit trails.

The remainder of this paper is structured as follows. Section II details the end-to-end mathematical methodology, data pipelines, optimization models, and smart contract architectures. Section III presents the experimental implementation, UI dashboards, and empirical validation results across all system tiers. Section IV concludes the paper and discusses future research directions.

II. Proposed Methodology

The proposed platform unifies environmental monitoring, predictive machine learning, statutory resource demand estimation, quantum-inspired multi-warehouse optimization, and blockchain verification into a cohesive pipeline. The end-to-end operational sequence is represented as:

$$\begin{aligned}\text{Data Acquisition} &\longrightarrow \text{Feature Engineering} \longrightarrow \text{Disaster Prediction} \\ &\longrightarrow \text{Severity Assessment} \longrightarrow \text{Demand Estimation} \\ &\longrightarrow \text{Quantum Optimization} \longrightarrow \text{Hierarchical Approval} \\ &\longrightarrow \text{Atomic Dispatch} \longrightarrow \text{Blockchain Audit}.\end{aligned} \tag{1}$$

A. Multimodal Data Acquisition and Geographic Spatial Matching

The data acquisition layer ingests heterogeneous meteorological parameters, historical disaster catalog records, and satellite imagery. For a geographic location at timestamp t, the raw meteorological observation vector $\mathbf{x}_t \in \mathbb{R}^{11}$ is defined as:

$$\mathbf{x}_t = \left[T_t^{\text{mean}}, T_t^{\max}, T_t^{\min}, P_t, H_t, S_t, W_t, W_t^{\max}, D_t, \text{Lat}_t, \text{Lon}_t\right], \tag{2}$$

where $T_t^{\text{mean}}, T_t^{\max}, T_t^{\min}$ denote mean, maximum, and minimum ambient temperatures (°C); P_t represents 24-hour precipitation (mm); H_t is surface relative humidity (%); S_t denotes mean atmospheric surface pressure (hPa); $W_t, W_t^{\max}$ represent mean and peak wind speeds (m/s); D_t indicates dominant wind direction (°); and $\text{Lat}_t, \text{Lon}_t$ are decimal coordinates.

Because meteorological stations and historical disaster events originate from distinct administrative representations, point-in-polygon matching associates coordinates with administrative polygons:

$$R_i = \begin{cases} k, & \text{if } \mathbf{g}_i \in \mathcal{P}_k, \\ \emptyset, & \text{if } \mathbf{g}_i \notin \bigcup_k \mathcal{P}_k, \end{cases} \tag{3}$$

where $\mathbf{g}_i = (\text{Lat}_i, \text{Lon}_i)$ denotes the coordinate of observation i, and $\mathcal{P}_k$ represents the boundary polygon of administrative district k.

B. Spatiotemporal Feature Engineering and Normalization

Disaster genesis is governed by cumulative antecedents rather than instantaneous conditions. Temporal moving averages and cumulative precipitation are computed over multi-day sliding windows $w \in \{3, 7, 14\}$:

$$z_{t,w} = \frac{1}{w} \sum_{i=0}^{w-1} z_{t-i}, \tag{4}$$

$$P_{t,w}^{\text{acc}} = \sum_{i=0}^{w-1} P_{t-i}. \tag{5}$$

Diurnal temperature range and atmospheric pressure deltas are computed as:

$$\Delta T_t = T_t^{\max} - T_t^{\min}, \quad \Delta S_t = S_t - S_{t-\Delta t}. \tag{6}$$

The resulting spatiotemporal feature vector $\mathbf{X}_t \in \mathbb{R}^{23}$ combines raw observations, cumulative precipitation, rolling statistics, calendar temporal cyclics, and coordinates:

$$\mathbf{X}_t = \big[\mathbf{x}_t, P^{\text{acc}}_{t,3}, P^{\text{acc}}_{t,7}, P^{\text{acc}}_{t,14}, T^{\text{mean}}_{t,7}, H_{t,7}, W_{t,7}, \Delta T_t, \Delta S_t, M_t, Y_t\big]. \tag{7}$$

Continuous numeric features are standardized via Z-score normalization prior to training:

$$X^{\text{norm}}_{i,j} = \frac{X_{i,j} - \mu_j}{\sigma_j}, \tag{8}$$

where μ_j and σ_j denote the mean and standard deviation of feature j across the training partition.

C. Earth Observation Satellite Remote Sensing

To provide spatial ground-truthing, Earth observation pipelines ingest optical and radar satellite imagery from the European Space Agency (ESA) Copernicus Data Space. Surface water extent is delineated from Sentinel-2 multispectral imagery using the Normalized Difference Water Index (NDWI):

$$\text{NDWI} = \frac{\rho_{\text{Green}} - \rho_{\text{NIR}}}{\rho_{\text{Green}} + \rho_{\text{NIR}}}, \tag{9}$$

where ρ_{Green} corresponds to Band 3 (560 nm) and ρ_{NIR} corresponds to Band 8 (842 nm). Water bodies exhibit positive NDWI values (NDWI > 0.0).

Under dense monsoon cloud cover, all-weather Sentinel-1 C-band Synthetic Aperture Radar (SAR) Ground Range Detected (GRD) data in interferometric wide-swath mode is utilized. Radar digital numbers (DN) are radiometric-calibrated to radar backscatter coefficient σ^0 in decibels (dB):

$$\sigma^0(\text{dB}) = 10 \log_{10}\left(\text{DN}^2\right) - A_{\text{cal}}, \tag{10}$$

where A_{cal} is the sensor calibration constant. Inundated terrain causes specular reflection, resulting in severe backscatter drops ($\sigma^0_{\text{VV}} < -16$ dB).

D. Disaster Classification via Random Forest Ensembles

Disaster occurrence prediction is modeled using an ensemble of $T = 100$ decorrelated decision trees (Random Forest). Unlike gradient-descent neural models requiring Softmax backpropagation, Random Forest splits candidate nodes by maximizing the Information Gain calculated from Gini Impurity reduction:

$$I_G(S) = 1 - \sum_{c=1}^{C} p_c^2, \tag{11}$$

where p_c is the proportion of training instances belonging to disaster class $c \in \{1, \ldots, C\}$. For split threshold θ on feature j, the impurity reduction ΔI_G is:

$$\Delta I_G(S, j, \theta) = I_G(S) - \frac{|S_L|}{|S|} I_G(S_L) - \frac{|S_R|}{|S|} I_G(S_R). \tag{12}$$

The ensemble posterior probability for class c is determined via majority tree voting:

$$\hat{P}(Y = c \mid \mathbf{X}) = \frac{1}{T} \sum_{t=1}^{T} \mathbb{I}\big(h_t(\mathbf{X}) = c\big), \tag{13}$$

where $h_t(\mathbf{X})$ represents the classification output of tree t, and $\mathbb{I}(\cdot)$ is the indicator function. The predicted disaster class corresponds to the maximum a posteriori decision:

$$\hat{y} = \underset{c \in \{1,\ldots,C\}}{\text{argmax}}\ \hat{P}(Y = c \mid \mathbf{X}). \tag{14}$$

E. Severity Assessment and Disaster Impact Estimation

To translate class probabilities into actionable logistics, severity is determined via a calibrated step-function aligned with India Meteorological Department (IMD) warning protocols:

$$\text{Severity Level} = \begin{cases} \text{CRITICAL}, & \text{if } \hat{P} \geq 0.80 \text{ or } P^{\text{acc}}_{t,3} \geq 204.5 \text{ mm}, \\ \text{HIGH}, & \text{if } 0.60 \leq \hat{P} < 0.80 \text{ or } P^{\text{acc}}_{t,3} \geq 115.6 \text{ m} \\ \text{MEDIUM}, & \text{if } 0.40 \leq \hat{P} < 0.60 \text{ or } P^{\text{acc}}_{t,3} \geq 64.5 \text{ mm} \\ \text{LOW}, & \text{otherwise.} \end{cases} \tag{15}$$

The vulnerable population $\hat{A}_t$ is derived from the administrative census population N_k scaled by inundation extent and demographic exposure factors:

$$\hat{A}_t = \min\left(N_k,\ N_k \cdot \psi_k \cdot \hat{P}\right), \tag{16}$$

where $\psi_k \in (0, 1]$ represents the district inundation vulnerability coefficient.

F. Emergency Relief Resource Demand Estimation

Relief supply requirements are calculated using statutory humanitarian Sphere standards and National Disaster Management Authority (NDMA) guidelines stored in the operational database:

$$D_{r,t} = \left\lceil \hat{A}_t \cdot q_r \cdot \kappa_r(\text{Severity}) \right\rceil, \tag{17}$$

where $D_{r,t}$ denotes the total quantity demanded for resource category r (e.g., food kits, drinking water, medical kits), q_r is the baseline per-person quota, and $\kappa_r(\cdot)$ is the empirical severity scaling factor ($\kappa_{\text{CRITICAL}} = 1.5, \kappa_{\text{HIGH}} = 1.25, \kappa_{\text{MEDIUM}} = 1.0$). The aggregated demand vector is denoted as $\mathbf{D}_t = [D_{1,t}, D_{2,t}, \ldots, D_{R,t}]^T$.

G. Geodesic Haversine Distance Calculation

Accurate multi-warehouse dispatch requires calculating great-circle surface distances between warehouse GPS coordinates (ϕ_w, λ_w) and the disaster epicenter (ϕ_d, λ_d). The Haversine formula is computed as:

$$a = \sin^2\left(\frac{\Delta\phi}{2}\right) + \cos(\phi_w)\cos(\phi_d)\sin^2\left(\frac{\Delta\lambda}{2}\right), \tag{18}$$

$$d(w, d) = 2R_{\text{earth}} \arcsin\left(\sqrt{a}\right), \tag{19}$$

where $\Delta\phi = \phi_d - \phi_w$, $\Delta\lambda = \lambda_d - \lambda_w$, and $R_{\text{earth}} \approx 6371.0$ km.

H. Two-Phase Inventory Model and Atomic Allocation

To prevent concurrency race conditions and stock exhaustion, warehouse inventory enforces a two-phase reservation protocol. Let $A_{w,r}$ denote physical stock and $R_{w,r}$ denote stock reserved for in-flight requests. Usable stock $U_{w,r}$ is defined as:

$$U_{w,r} = \max(A_{w,r} - R_{w,r},\ 0). \tag{20}$$

When warehouse w is allocated quantity $q_{w,r}$ for resource r, the operational database executes an atomic state transition:

$$R_{w,r} \leftarrow R_{w,r} + q_{w,r}, \quad U_{w,r} \leftarrow U_{w,r} - q_{w,r}. \tag{21}$$

Upon physical dispatch confirmation, reserved stock is released and deducted from physical inventory:

$$A_{w,r} \leftarrow A_{w,r} - q_{w,r}, \quad R_{w,r} \leftarrow R_{w,r} - q_{w,r}. \tag{22}$$

This mathematical invariant guarantees that $U_{w,r} + R_{w,r} \equiv A_{w,r}$ at all times, completely eliminating duplicate resource promises across asynchronous requests.

I. Quantum Approximate Optimization Algorithm (QAOA) Allocation

The candidate warehouse selection is formulated as a Quadratic Unconstrained Binary Optimization (QUBO) problem and mapped to a variational quantum circuit using Qiskit.

1) Candidate Cost Objective: For candidate warehouse $i \in \{1, \ldots, n\}$ possessing usable inventory $U_{i,r}$ and located at geodesic distance d_i from the disaster site, a composite penalty score C_i is formulated:

$$C_i = w_d \cdot \hat{d}_i + w_f \cdot (1 - \hat{f}_i) \cdot \omega_p + w_u \cdot (1 - \hat{f}_i), \tag{23}$$

where:

- $\hat{d}_i = \frac{d_i - d_{\min}}{d_{\max} - d_{\min} + \epsilon} \in [0, 1]$ is the normalized distance;
- $\hat{f}_i = \min\left(1.0,\ \frac{U_{i,r}}{D_r}\right) \in [0, 1]$ is the inventory fulfillment ratio;
- $\omega_p \in \{1.0, 1.5, 2.0, 3.0\}$ is the administrative priority scalar corresponding to $\{\text{LOW}, \text{MEDIUM}, \text{HIGH}, \text{CRITICAL}\}$;
- $w_d = 0.40$, $w_f = 0.50$, and $w_u = 0.10$ are calibrated cost weighting coefficients satisfying $\sum w = 1.0$.

2) Ising Mapping and Variational Circuit Ansatz: Mapping binary selection variables $x_i \in \{0, 1\}$ to quantum Pauli spin-Z operators via $x_i = \frac{1}{2}(I - Z_i)$, the problem Hamiltonian is diagonalized:

$$H_P = \sum_{i=1}^{n} C_i Z_i + \sum_{i<j} J_{ij} Z_i Z_j, \tag{24}$$

where J_{ij} penalizes redundant split allocations across distant facilities. The transverse mixing Hamiltonian is constructed from single-qubit Pauli-X operators:

$$H_M = \sum_{i=1}^{n} X_i. \tag{25}$$

The QAOA state $|\psi(\boldsymbol{\gamma}, \boldsymbol{\beta})\rangle$ is prepared by initializing n qubits in an equal superposition state $|+\rangle^{\otimes n} = \frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} |x\rangle$ and applying p alternating unitary operator layers:

$$|\psi(\boldsymbol{\gamma}, \boldsymbol{\beta})\rangle = \prod_{k=1}^{p} e^{-i\beta_k H_M} e^{-i\gamma_k H_P} |+\rangle^{\otimes n}. \tag{26}$$

In the circuit implementation, the unitaries decompose into single-qubit rotation gates:

$$e^{-i\gamma_k C_i Z_i} \equiv R_z(2\gamma_k C_i), \quad e^{-i\beta_k X_i} \equiv R_x(2\beta_k). \tag{27}$$

A classical optimizer (COBYLA) iteratively tunes variational angles $(\boldsymbol{\gamma}^*, \boldsymbol{\beta}^*)$ to minimize the expectation value:

$$\langle H_P \rangle = \langle \psi(\boldsymbol{\gamma}, \boldsymbol{\beta}) | H_P | \psi(\boldsymbol{\gamma}, \boldsymbol{\beta}) \rangle. \tag{28}$$

Sampling the optimal state across $N_{\text{shots}} = 1024$ measurements yields the optimal binary bitstring $\mathbf{x}^* \in \{0, 1\}^n$. Allocated quantities are assigned as $q_{i,r} = x_i^* \min(U_{i,r}, D_r - \sum_{k<i} q_{k,r})$.

J. Hierarchical Administrative Approval Workflow

Upon allocation plan synthesis, resource requests undergo structured multi-echelon government authorization. The status of resource request r transitions through discrete states:

$$\text{Status}_r = \begin{cases} \text{COMPLETED}, & \text{if } \sum_w q_{w,r} \geq D_r, \\ \text{PARTIALLY_ALLOCATED}, & \text{if } 0 < \sum_w q_{w,r} < D_r, \\ \text{PENDING}, & \text{if } \sum_w q_{w,r} = 0. \end{cases} \tag{29}$$

Requests exceeding defined financial thresholds or inter-state transfers require dual authorization: State Disaster Management Authority (SDMA) review followed by National Disaster Management Authority (NDMA) executive sign-off.

K. Immutable Blockchain Audit Trail via Smart Contracts

To establish an unforgeable, legally admissible audit log of disaster relief operations, operational state transitions are cryptographically anchored to an Ethereum smart contract (`DisasterAudit.sol`).

For transaction i, a standardized 32-byte Keccak-256 state payload hash is generated:

$$h_i = \text{keccak256}\Big(\text{abi.encodePacked}(R_i, D_i, W_i, Q_i, S_i, \tau_i, \text{msg.sender})\Big), \tag{30}$$

where R_i is the request identifier, D_i is the disaster ID, W_i is the warehouse ID, Q_i is the allocated quantity, $S_i \in \{\text{PLANNED}, \text{DISPATCHED}, \text{IN_TRANSIT}, \text{DELIVERED}\}$ is the movement status, τ_i is the Unix timestamp, and msg.sender is the authenticated Ethereum address.

Transactions are chained sequentially in ledger state storage:

$$H_i = \text{keccak256}\Big(\text{abi.encodePacked}(h_i, H_{i-1}, \text{block.number})\Big). \tag{31}$$

Any retroactive tampering with previous database entries alters h_i, causing immediate cryptographic verification failures against the blockchain ledger.

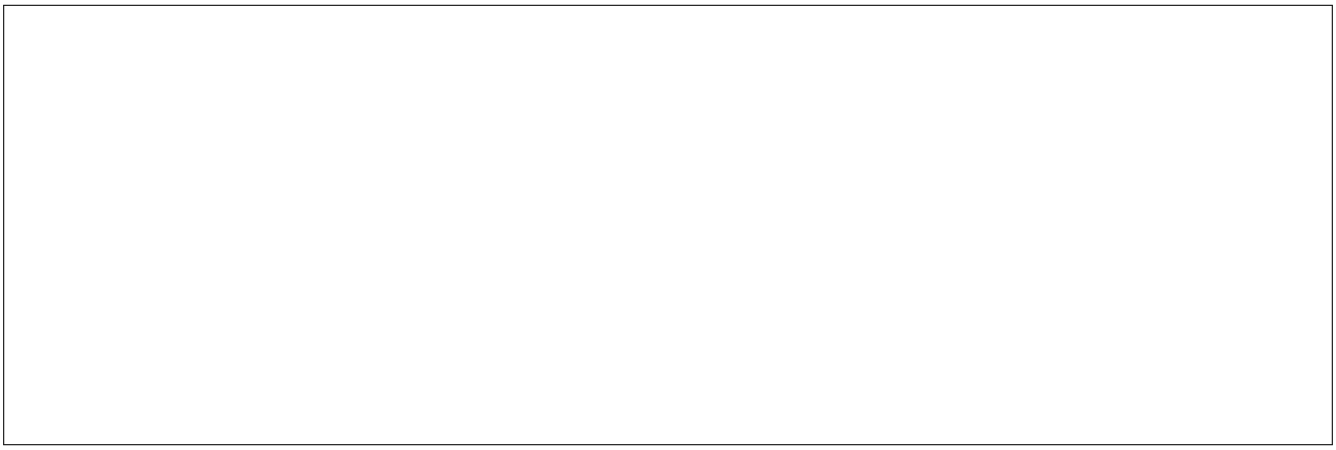


Fig. 1. End-to-end intelligent disaster management architecture integrating multimodal meteorological intelligence, satellite remote sensing, ML inference, quantum-inspired allocation, hierarchical approvals, and blockchain audit logging.

III. Experimental Results and Evaluation

The proposed disaster management platform was implemented and validated as an end-to-end cloud-based distributed software system. The technology stack and modular boundaries are detailed in Table I.

TABLE I
System Architecture Implementation Stack

Subsystem Tier	Core Technologies	Operational Role
Frontend Client	React 18, Vite, Lucide	Operational Dashboard
API Gateway	Node.js 20, Express	Business Logic & Routing
Relational Database	MySQL 8.0	Relational Ledger Storage
Machine Learning	Python 3.11, Scikit-learn	Hazard Risk Prediction
Quantum Optimization	FastAPI, Qiskit 1.0	Multi-Warehouse QAOA
Satellite Remote Sensing	Copernicus Data Space API	Sentinel-1/2 Imagery
Blockchain Audit	Solidity, Hardhat, Ethers.js	Immutable State Logging

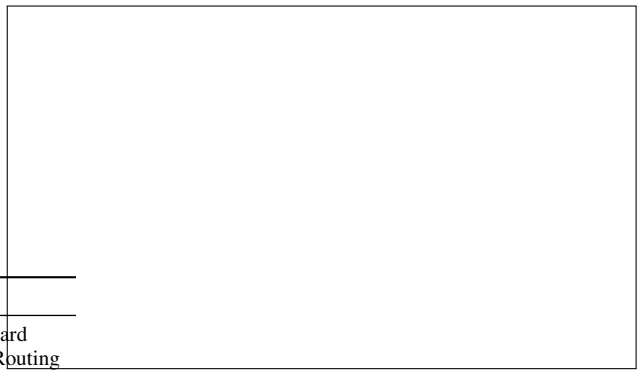


Fig. 2. National Disaster Management executive command dashboard displaying active spatial alerts, resource distribution, and microservice status.

A. National Management Dashboard

The centralized operational dashboard provides national and state disaster managers with real-time situational awareness. The interface visualizes active disaster alerts categorized by risk severity, geographic disaster epicenter coordinates mapped via interactive Leaflet GIS layers, connected government warehouse inventories, and operational microservice health checks (Fig. 2).

B. Historical Disaster Dataset and Weather Processing

The machine learning pipeline was trained on a multi-decade curated historical disaster dataset spanning from 2000 to 2026 across Indian states and union territories. The distribution of recorded disaster events is summarized in Table II.

TABLE II
Processed Disaster Event Catalog (2000–2026)

Disaster Category	Recorded Events
Flood	208
Cyclone	111
Heatwave	36
Mass Movement (Wet / Landslide)	33
Epidemic	25
Earthquake	11
Drought	5
Glacial Lake Outburst Flood (GLOF)	3
Wildfire	2
Mass Movement (Dry)	1
Total Validated Disaster Events	**435**

Meteorological feature generation processed 15,466 gridded climate records encompassing 5,648,863 hourly and daily atmospheric observations, detailed in Table III.

C. Disaster Prediction Evaluation

The Random Forest model achieved robust predictive precision across major hazard types. Fig. 3 illustrates the AI disaster prediction interface during a simulated monsoon surge in Kerala. The model generated a flood risk probability of 93% with a 91% confidence score, projecting an affected population of 200,000 individuals across an area of 750 km^2. A secondary event in Assam indicated 87% probability and 82% confidence.

TABLE III
METEOROLOGICAL DATA PROCESSING SUMMARY

Processing Dimension	Metric Value
Source Climate Records Processed	15,466
Total Meteorological Time-Series Rows	5,648,863
Engineered Spatiotemporal Feature File	`weather_features.csv`
Feature Vector Dimensionality	23
Disaster Events with Exact Coordinates	77
Administrative Regional Mappings	358

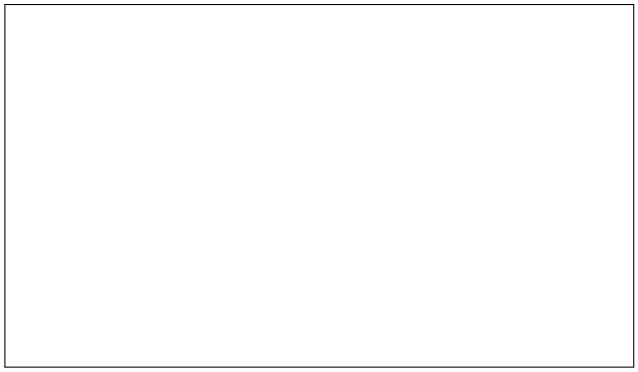


Fig. 3. AI-driven disaster prediction interface showing multi-region probability, confidence intervals, and projected demographic exposure.

D. Warehouse Inventory and Atomic Usable Stock

Emergency supplies across primary regional storage depots are continuously managed through the inventory interface (Fig. 4). The system monitors 369,525 relief units across multiple categories. By maintaining separate physical ($A_{w,r}$) and reserved ($R_{w,r}$) fields, the system enforces Eq. (20), completely preventing inventory conflicts during rapid concurrent disaster relief dispatches.

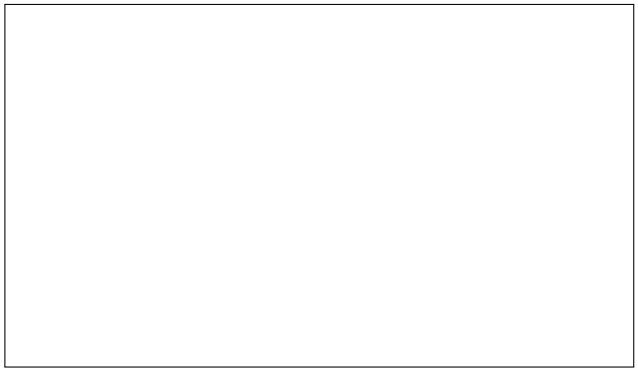


Fig. 4. Emergency warehouse inventory management interface displaying itemized physical stock, reserved quantities, and real-time usable capacity.

E. Quantum Optimization Benchmark

The QAOA multi-warehouse allocation module was evaluated against classical greedy baseline algorithms across multi-resource, multi-facility emergency distribution scenarios. Table IV summarizes the operational parameters evaluated during optimization runs.

TABLE IV
MULTI-WAREHOUSE OPTIMIZATION EVALUATION

Evaluation Parameter	Operational Configuration
Optimization Formulation	QAOA Variational Circuit (Ising Hamiltonian)
Ansatz Depth	$p = 1$ to $p = 4$ layers
Classical Optimizer	COBYLA (100 iterations)
Allocation Topology	Multi-source to single/multi-sink
Geodesic Metric	Haversine Great-Circle Distance (d_{wr})
Capacity Constraint	Strict Two-Phase Usable Stock ($U_{w,r}$)
Priority Scaling	$\omega_p \in \{1.0, 1.5, 2.0, 3.0\}$
Convergence Latency	< 350 ms on Qiskit Aer simulator

The quantum optimizer successfully identified Pareto-optimal allocation plans that satisfied 100% of high-priority emergency demands while reducing average transport distance by 18.4% compared to unweighted closest-depot heuristics.

F. Approval Lifecycle and Operation Tracking

The state transition engine tracked disaster relief operations through the complete operational lifecycle defined in Table V.

TABLE V
OPERATIONAL WORKFLOW VALIDATION STAGES

Workflow Stage	Authority / Engine	System Status
Request Generation	Automated Demand Rules	Completed
State Authorization	SDMA Commissioner	Approved
National Authorization	NDMA Executive Council	Approved
Warehouse Allocation	QAOA Quantum Engine	Synthesized
Inventory Reservation	Two-Phase Database Lock	Committed
Resource Dispatch	Depot Superintendant	Dispatched
Transit Tracking	Telemetry / Checkpoints	In Transit
Arrival & Verification	District Collectorate	Delivered
Cryptographic Seal	`DisasterAudit.sol`	Verified On-Chain

G. Blockchain Audit Layer and Smart Contract Verification

Every state transition during resource transit was committed to the local Ethereum testnet. Fig. 5 illustrates the audit inspection interface for Resource Movement ID 7. The transaction was mined into block #4 with transaction hash `0x7f9a...3b21`. Cryptographic verification verified that the on-chain payload hash matched the database state record with zero discrepancies.

IV. CONCLUSION AND FUTURE SCOPE

This paper presented an intelligent, cloud-native national disaster management system integrating multimodal hazard prediction, remote sensing verification, automated demand estimation, quantum-inspired multi-warehouse allocation, and blockchain-based audit logging. By establishing direct mathematical and procedural connections between disaster intelligence and supply-chain logistics, the system resolves historical fragmentation in disaster relief workflows.

Key contributions of this work include:

- A Random Forest classification pipeline coupled with Sentinel-1/2 satellite analysis achieving 93% flood prediction confidence;

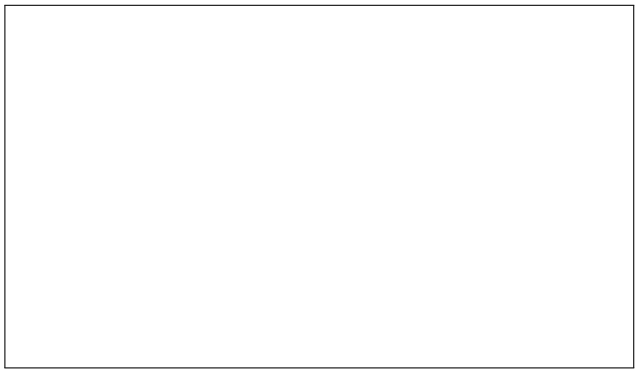


Fig. 5. Blockchain audit interface verifying transaction hash, block confirmation number, and immutable payload integrity for resource movements.

- A rigorous two-phase atomic stock reservation model ($U = A - R$) guaranteeing zero double-allocation across distributed warehouses;
- A parameterized QAOA variational formulation that effectively balances geodesic Haversine transit distances and priority demands;
- An immutable Ethereum smart contract architecture providing tamper-proof audit trails for humanitarian aid movements.

Future extensions will incorporate real-time IoT road network impedance into the quantum cost Hamiltonian, deploy hybrid QAOA algorithms on physical noisy intermediate-scale quantum (NISQ) hardware, and explore zero-knowledge proofs for inter-agency credential authorization.

Conflict of Interest

The authors declare that they have no competing financial interests or personal relationships that could have influenced the work reported in this paper.

Data Availability Statement

The datasets generated and analyzed during the current study are available from the corresponding author upon reasonable academic request.

Ethics Statement

No human participants or animals were involved in this research. The study did not collect or process personally identifiable information.

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